NLP Preprocessing Practice Questions

1. Corpus

You are building a sentiment analysis model for movie reviews.
Question:

- What is a corpus?
- Is the IMDB movie review dataset considered a corpus? Why?

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2. Lowercasing

Consider the text:
PHITRON is Amazing! I love studying NLP at PHITRON.

Question:

- Why do we perform lowercasing?
- What will be the output after lowercasing?
- How does lowercasing help reduce vocabulary size?

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3. HTML Tag Removal

Consider the text:
<p>Welcome to PHITRON!</p>

Question:

- Which regex pattern can remove HTML tags?
- Explain why the pattern <.*?> works.
- What is the purpose of ? in this regex?

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4. URL Removal

Consider the text:
For more details visit https://phitron.io or www.phitron.io

Question:

- Explain the regex:

r'https?://\S+|www\.\S+'

- What does:
    - https?
    - \S+
    - www\.

      mean?

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5. Punctuation Removal

Consider the code:
text.translate(str.maketrans('', '', string.punctuation))

Question:
- What is the role of string.punctuation?
- Why are the first two arguments of maketrans() empty strings?
- What would happen if punctuation is not removed before tokenization?

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6. Stopword Removal

Consider the sentence:
I am studying Machine Learning in the university.

Question:
- What are stopwords?
- Which words would likely be removed?
- Why do we convert stopwords into a set instead of keeping them as a list?

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7. Tokenization

Consider the sentence:
I am going to visit Dhaka!

Question:
- What is tokenization?
- What is the difference between:

text.split()

and
word_tokenize(text)

- Which one handles punctuation better?

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8. Stemming

Consider the words:
running, studies, playing, happiness

Question:

- What is stemming?
- Apply stemming conceptually to each word.
- Why might stemming produce words that are not found in a dictionary?

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9. Lemmatization

Consider the words:
running, children, better, studies

Question:

- What is lemmatization?
- What would be the lemma of each word?
- Why is lemmatization generally more accurate than stemming?

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10. Complete NLP Pipeline

Consider the text:
OMG!!! I was studying NLP at PHITRON 😊.
Visit https://phitron.io for more details.

Question:
Describe the preprocessing steps in order and show the expected output after each stage:

1. Lowercasing
2. URL removal
3. Emoji removal
4. Punctuation removal
5. Tokenization
6. Stopword removal
7. Stemming
8. Lemmatization